AI Workforce Odyssey

Design Choices & Rationale

The following table documents every significant design choice made during the model design process, along with the reasoning behind each decision. These choices were developed iteratively through multiple design revisions (v1 → v2 → v3 → v3.1 → v3.2), incorporating external feedback at each stage. Version 3.2 reflects final parameter tuning based on test runs of both scenarios.

#	Design Choice	Rationale
1	**30 workers**	Maximum allowed by the assignment. More workers produce richer emergent dynamics and more data points for analysis, while still keeping the model small and readable.
2	**4 task types: Routine-Cognitive, Non-Routine Cognitive, Relationship-Based, Non-Routine Manual**	Captures a realistic spectrum of AI exposure. Routine-Cognitive faces the highest disruption risk; Relationship-Based is a hybrid category reflecting jobs like sales and partnerships where routine elements (CRM, reporting) coexist with deeply non-routine relationship work.
3	**Routine-share as the primary risk driver (0–1 scale)**	A worker's percentage of routine tasks determines how vulnerable they are to AI automation. This is grounded in task-based labor economics and mirrors the AI-exposure scores used in the Stanford Digital Economy Lab research.
4	**3 seniority levels: Junior, Mid, Senior**	Junior workers face higher disruption risk due to less institutional protection and less specialized expertise. This reflects the observed pattern that entry-level workers are disproportionately affected by AI-driven employment changes.
5	**2 behavioral traits: Adaptability and Resilience**	Adaptability governs whether a worker can enter and benefit from training. Resilience determines whether they disengage under pressure (becoming disgruntled). Two traits produce meaningful individual variation without overcomplicating the model.
6	**5 states: Employed, At-Risk, In-Training, Unemployed, Recovered**	Enough states to capture a realistic workforce transition pipeline, but few enough to keep the model readable and the memo explainable. Earlier versions had 7 states; this was simplified based on feedback that Augmented and Disgruntled work better as attributes than standalone states.
7	**Recovered with subtypes (re-employed vs augmented) rather than separate states**	Keeps the state space clean while still tracking an important outcome difference. Augmented workers represent those who successfully adapted to work alongside AI; re-employed workers returned to similar roles.
8	**Disgruntled as a flag, not a state**	Workers who disengage under pressure are still technically at-risk, not in a separate labor market state. The flag modifies their behavior (reduces effective adaptability by 20) without adding state complexity.
9	**Accelerating automation pressure**	Pressure starts low and increases each tick, mirroring the Stanford finding that AI-related employment effects became significant in 2024 and are growing without reversal. This produces staggered disruption rather than a single shock.

10	**Disruption-risk formula normalized to 0–1**	All components (routine-share, automation pressure, seniority penalty) are normalized so risk stays bounded throughout the simulation. Without this, risk values would explode numerically and cause mass disruption too early.
11	**Primary mechanism: Unequal adaptation, not seat scarcity**	The core story is that equal training access does not produce equal recovery. Workers diverge because of trait differences, timing, and peer effects—not because they can't get a seat. This is more realistic than a pure bottleneck model.
12	**Training seats as a secondary constraint (8 Tech-Driven / 20 Human-Centric)**	Seats are calibrated to be non-binding most of the time, only occasionally constraining during concentrated disruption waves. This avoids making queue competition the dominant mechanism while preserving it as an amplifier under stress.
13	**Scarring effect as a tunable slider (0.5/tick Tech-Driven, 0.3/tick Human-Centric)**	Creates a feedback loop where long unemployment makes re-entry progressively harder. Originally hardcoded at 1.0/tick, but testing showed this made unemployment permanent. Reduced to 0.5 and made a slider parameter so it can be tuned without editing code. Human-Centric uses 0.3 to reflect that better support systems reduce scarring.
14	**Peer influence with spatial radius**	Workers near recovered or in-training peers get a motivation boost (+5 per neighbor, capped at 20). Workers are placed randomly and never move, so any recovery or stagnation clusters that form are genuinely emergent from local interactions.
15	**Disgruntled contagion: −2 resilience per tick near 2+ disgruntled workers**	At-risk workers near multiple disgruntled peers have their resilience eroded, potentially becoming disgruntled themselves. This creates contagion pockets that are not directly programmed as an outcome.
16	**Recovered workers can return to at-risk (threshold + 0.15 buffer)**	Prevents Recovered from being a permanent absorbing state. The buffer means recovered workers are more resilient than first-timers but not immune to very high late-simulation pressure. Adds realism without disrupting the core dynamics.
17	**Random spatial placement, no movement**	Workers are positioned randomly at setup and stay fixed. This ensures that any spatial clustering of outcomes is emergent, not an artifact of engineered placement. Earlier versions grouped workers by state, which would have hard-coded clustering.
18	**Same model structure for both scenarios, only parameter values change**	This is the correct experimental design for scenario comparison. Both scenarios use identical code and rules; the only differences are slider settings. Any outcome differences are therefore attributable to the parameter changes, not structural differences.
19	**Tech-Driven vs Human-Centric scenario parameters tuned through testing**	Initial parameters produced a Tech-Driven scenario that was too harsh (permanent unemployment traps) and a Human-Centric scenario that was too gentle (most workers never disrupted). Parameters were adjusted after test runs: Tech-Driven threshold raised to 0.50, augmentation to 30%; Human-Centric threshold lowered to 0.55, acceleration raised to 0.20. This ensures both scenarios produce meaningful dynamics.

20	**3 plots and monitors tracked across both scenarios**	Unemployment over time, state counts over time, and recovery by task type provide dynamic comparison. Monitors track share recovered, average time unemployed, augmentation rate, and seats available. Plots were trimmed from an initial 5 to 3 to keep the interface clean and focused on the outputs that matter most for the memo figures.
21	**Scenario preset buttons in the interface**	One-click buttons that set all 10 slider parameters to either Tech-Driven or Human-Centric values. Ensures exact reproducibility between runs and eliminates manual slider-setting errors, making the scenario comparison cleaner and more reliable.
22	**All parameters exposed as sliders (no hardcoded values)**	Every tunable parameter including scarring-rate is a slider. This means the model can be fully explored and adjusted without editing code — a deliberate design choice to support iterative testing and transparency about what drives outcomes.
23	**Parameter tuning driven by test runs, not theory alone**	Initial parameter values were based on design logic. After running both scenarios, several values were adjusted to produce more balanced and realistic dynamics. This iterative calibration process is documented and reflects standard ABM practice.